THE SOLUTIONS

1. (a) Two matrices are given as

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 3 \\ 1 & \alpha & 1 \end{pmatrix}, \quad \text{and} \quad B = \begin{pmatrix} 3 & 2 & 4 \\ 3 & 2 & -1 \\ 1 & 1 & -1 \end{pmatrix}$$

(i) Let $\alpha = 1$. Find $\det(A)$, as well as $\det(B)$ and $\det(AB)$. Finally, use Gauss-Jordan elimination to obtain A^{-1}. [6]

(ii) Let $\alpha = 3$. Give conditions on $\underline{\mathbf{b}} = (2, \beta, 1)^T$ so that the matrix equation $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ has no solutions, or infinitely many solutions. Can the system have a unique solution? Either way, explain why. Finally, for the case of infinitely many solutions, find these and give a geometric interpretation. [6]

Solution (i) For A, expand by third row:

$\det(A) = 1(6-2) - \alpha(3-2) + 1(1-2) = -\alpha + 3 = 2$, when $\alpha = 1$.

For B use a row operation: $R_1 - R_2$, to write

$$\det(B) = \begin{vmatrix} 0 & 0 & 5 \\ 3 & 2 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 5(3-2) = 5 \, .$$

Hence $\det(AB) = \det(A)\det(B) = 2(5) = 10$.

Almost all done correctly. A few people (less than on previous occasions) multiplied to get AB and then found the determinant, sometimes correctly.

For Gauss-Jordan, augment A with the identity matrix, and use row operations to make the left-half into the identity: $(A : I)$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \underset{R_3 - R_1}{\overset{R_2 - R_1}{\sim}} \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array}\right) \underset{-R_2}{\sim} \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array}\right)$$

$$\underset{R_3 + R_2}{\overset{R_1 - 2R_2}{\sim}} \left(\begin{array}{ccc|ccc} 1 & 0 & 4 & -1 & 2 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & -2 & 0 & -1 & 1 \end{array}\right) \underset{-\frac{1}{2}R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 4 & -1 & 2 & 0 \\ 0 & 1 & -1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} \end{array}\right) \underset{R_2 + R_3}{\overset{R_1 - 4R_3}{\sim}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & 2 \\ 0 & 1 & 0 & 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} \end{array}\right)$$

so that $A^{-1} = \begin{pmatrix} -1 & 0 & 2 \\ 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$.

Some arithmetic errors, but mostly done well.

(ii) We now have $\det(A) = 0$, making the matrix singular, so the system can never have a unique solution, as this is only possible when A is invertible. Use Gaussian elimination on the augmented coefficient matrix:

$$(A : \underline{\mathbf{b}}) = \left(\begin{array}{ccc:c} 1 & 2 & 2 & 2 \\ 1 & 1 & 3 & \beta \\ 1 & 3 & 1 & 1 \end{array}\right) \underset{\substack{R_2-R_1 \\ R_3-R_1}}{\sim} \left(\begin{array}{ccc:c} 1 & 2 & 2 & 2 \\ 0 & -1 & 1 & \beta-2 \\ 0 & 1 & -1 & -1 \end{array}\right) \underset{R_3+2R_2}{\sim} \left(\begin{array}{ccc:c} 1 & 2 & 2 & 2 \\ 0 & -1 & 1 & \beta-2 \\ 0 & 0 & 0 & \beta-3 \end{array}\right)$$

and we have echelon form. If $\beta \neq 3$, there is a pivot in the right-most column of the echelon form and the system in inconsistent, so no solutions, unless $\beta = 3$. For $\beta = 3$, we have infinitely many solutions. We now have

$$(A : \underline{\mathbf{b}}) \sim \left(\begin{array}{ccc:c} 1 & 2 & 2 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Row two has $-y+z=1$. Let $z = \lambda \in \mathbb{R}$, a free variable. Then $y = -1+\lambda$ and row one has $x = 2-2y-2z = 2-2(-1+\lambda)-2\lambda = 4-4\lambda$. In vector form this is

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -4 \\ 1 \\ 1 \end{pmatrix},$$

the vector equation of a line in $\mathbb{R}^3$.

Mostly done well, a few people got the correct vector equation, but forgot to talk about conditions regarding unique/no/infinite solutions, and vice-versa.

(b) (i) Use the definition of the Laplace transform to prove the first shift theorem:

$$\mathscr{L}[e^{at}f(t)] \;=\; \bar{f}(s-a)\;,$$

where $\mathscr{L}[f(t)] \;=\; \bar{f}(s)$. State any necessary condition. [4]

Solution: Using the integral definition of the Laplace Transform,

$$\mathscr{L}\left[e^{at}f(t)\right] \;=\; \int_0^\infty e^{at}f(t)e^{-st}\,dt \;=\; \int_0^\infty f(t)e^{-(s-a)t}\,dt = \overline{f}(s-a)\,.$$

For convergence, we require $\mathrm{Re}(s-a) > 0$. For the transform $\bar{f}(s)$ to exist, we also require $\lim_{t\to\infty} f(t)e^{-st} \;=$ 0.

Almost everyone got this right, but most people didn't include a condition.

(ii) Use the first shift theorem to obtain the inverse Laplace Transform of

$$\frac{5}{(s+3)(s^2+4s+13)}\;.$$

[5]

Solution: First use partial fractions:

$$\frac{5}{(s+3)(s^2+4s+13)} \;=\; \frac{A}{s+3}+\frac{Bs+C}{s^2+4s+13} \Rightarrow 5 = A(s^2+4s+13)+(Bs+C)(s+3)\;,$$

and letting $s=-3$, we get $A=\dfrac{1}{2}$, letting $s=0$ we get $C=-\dfrac{1}{2}$. Any other value will do. Let $s=1$, and obtain $B=-\dfrac{1}{2}$. Hence

$$\frac{5}{(s+3)(s^2+4s+13)} \;=\; \frac{1}{2}\left[\frac{1}{s+3}-\frac{s+1}{s^2+4s+13}\right] \;=\; \frac{1}{2}\left[\frac{1}{s+3}-\frac{s+2}{(s+2)^2+9}+\frac{1}{(s+2)^2+9}\right]\;,$$

where the last two terms are arranged to apply the first shift theorem:

$$\begin{aligned}\frac{1}{2}\mathscr{L}^{-1}\left[\frac{1}{s+3}-\frac{s+2}{(s+2)^2+9}+\frac{1}{(s+2)^2+9}\right] \;&=\; \frac{1}{2}\left(e^{-3t}+e^{-2t}\mathscr{L}^{-1}\left[-\frac{s}{s^2+9}+\frac{1}{s^2+9}\right]\right)\\ &=\; \frac{1}{2}\left[e^{-3t}+e^{-2t}\left(\frac{1}{3}\sin(3t)-\cos(3t)\right)\right]\;.\end{aligned}$$

A lot of correct answers. Most common mistake: not shifting every occurence of $s+2 \to s$, i.e. something like taking the inverse LT of $\dfrac{s}{(s+2)^2+9}+\dfrac{1}{(s+2)^2+9}$ *and getting* $e^{-2t}\left(\cos(3t)+\sin(3t)\right)$ *or similar.*

(iii) Use Laplace transforms and (ii) to solve the Ordinary Differential Equation

$$\frac{d^2x}{dt^2}+4\frac{dx}{dt}+13x = 5e^{-3t}$$

which satisfies the initial conditions $x(0)=1$ and $x'(0)=-2$. [4]

Solution: Taking Laplace Transforms of the ODE we get

$$s^2\bar{x} - sx(0) - x'(0) + 4[s\bar{x} - x(0)] + 13\bar{x} = \frac{5}{s+3} \Rightarrow (s^2+4s+13)\bar{x} - s - 2 = \frac{5}{s+3}$$

$$\Rightarrow \bar{x} = \frac{5}{(s+3)(s^2+4s+13)} + \frac{s+2}{s^2+4s+13}\,.$$

Hence using (ii) we have

$$x(t) = \mathscr{L}^{-1}[\bar{x}] = \frac{1}{2}\left[e^{-3t} + e^{-2t}\left(\frac{1}{3}\sin(3t) - \cos(3t)\right)\right] + e^{-2t}\cos(3t) = \frac{1}{2}e^{-3t} + \frac{1}{6}e^{-2t}\sin(3t) + \frac{1}{2}e^{-2t}\cos(3t)\,.$$

Mostly done well; most people who implemented the initial conditions correctly to get the $s+2$ term on the RHS finished with the right form.

2. (a) Obtain $F(\omega)$, the Fourier transform of a real signal $f(t)$, defined as

$$f(t) \;=\; \begin{cases} 0 & \text{for } t<0\,, \\ e^{-t}\cos(t) & \text{for } t\geq 0\,. \end{cases}$$

Show that we can write

$$F(\omega) \;=\; \frac{1+i\omega}{(1+i\omega)^2+k}\,,$$

and provide the value of the real number k. [7]

Solution: We let $\cos(t) \;=\; \frac{1}{2}\left(e^{it}+e^{-it}\right)$ and rewrite the transform as

$$F(\omega) \;=\; \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\,dt \;=\; \frac{1}{2}\int_0^{\infty} e^{-t}\left(e^{it}+e^{-it}\right)e^{-i\omega t}\,dt \;=\; \frac{1}{2}\int_0^{\infty} e^{it}e^{-(1+i\omega)t}\,dt+\frac{1}{2}\int_0^{\infty} e^{-it}e^{-(1+i\omega)t}\,dt$$

and apply the frequency shift property (Formula sheet) to get $\frac{1}{2}G(\omega\pm 1)$ where

$$G(\omega) \;=\; \int_0^{\infty} e^{-(1+i\omega)t}\,dt \;=\; \left[\frac{e^{-(1+i\omega)t}}{-(1+i\omega)}\right]_0^{\infty} \;=\; \frac{1}{1+i\omega}\,,$$

given that $\lim_{t\to\infty} e^{-t}=0$. Hence

$$F(\omega)=\frac{1}{2}\left(\frac{1}{1+i(\omega+1)}+\frac{1}{1+i(\omega-1)}\right) \;=\; \frac{1}{2}\left[\frac{1}{(1+i\omega)+i}+\frac{1}{(1+i\omega)-i}\right] \;=\; \frac{1+i\omega}{(1+i\omega)^2+1}\,,$$

as required, with $k=1$.

Slightly longer, we could also integrate directly, and take $F(\omega)=$

$$=\frac{1}{2}\int_0^{\infty} e^{-t}\left(e^{it}+e^{-it}\right)e^{-i\omega t}\,dt \;=\; \frac{1}{2}\int_0^{\infty} e^{(-1+i-i\omega)t}+e^{(-1-i-i\omega)t}\,dt \;=\; \frac{1}{2}\left[\frac{e^{(-1+i-i\omega)t}}{-1+i-i\omega}+\frac{e^{(-1-i-i\omega)t}}{-1-i-i\omega}\right]_0^{\infty}\,,$$

and arrive at the same result.

A large variety of attempted methods, often successful. The second version above was the most common one. Lots of people got lost somewhere or other and attempted to force the expected outcome. Other ways included integrating by parts twice, beginning with

$$\int_0^{\infty} e^{-t}\,\cos t\;e^{-i\omega t}\,dt$$

but some taking $u=\cos t, dv=e^{-(1+i\omega)t}\,dt$ and others taking $u=e^{-(1+i\omega)t}, dv=\cos t\,dt$, and some even taking $u=\cos t\;e^{-t}, dv=e^{-i\omega t}\,dt$. Some wrote it as a Laplace transform, letting $s=1+i\omega$ to get

$$\int_0^{\infty} \cos t\;e^{-st}\,dt \;=\; \mathscr{L}[\cos t] \;=\; \frac{s}{s^2+1},$$

getting the correct answer. Others letting $s=i\omega$ and then calculating $\mathscr{L}[e^{-t}\cos t]$ with the first shift-theorem. Amazing!

(b) A function of two variables is given as

$$f(x,y) \;=\; x\cos y + y\sin x\,.$$

(i) Obtain the Taylor series expansion of $f(x,y)$, around the point $(\pi/2,0)$, to second order. [5]

(ii) Hence find the equation of the tangent plane to the surface $z = f(x,y)$ at the point $(\pi/2,0)$. [4]

Solution: (i) Begin with $f(\pi/2,0) = \dfrac{\pi}{2}$ and obtain first and second derivatives:
$f_x = \cos y + y\cos x \Rightarrow f_x(\pi/2,0) = 1$,
$f_y = -x\sin y + \sin x \Rightarrow f_y(\pi/2,0) = 1$,
$f_{xx} = -y\sin x \Rightarrow f_{xx}(\pi/2,0) = 0$,
$f_{xy} = -\sin y + \cos x \Rightarrow f_{xy}(\pi/2,0) = 0$,
$f_{yy} = -x\cos y \Rightarrow f_{yy}(\pi/2,0) = -\dfrac{\pi}{2}$.

Hence to second order, the formula sheet gives the Taylor expansion
$f(x,y) \approx f + f_x h + f_y k + \dfrac{1}{2}\left(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2\right)$
where on the rhs, f and all derivatives are evaluated at $(\pi/2,0)$, $h = x - \pi/2$ and $k = y - 0$. Finally, the Taylor series is

$$f(x,y) \;\approx\; \frac{\pi}{2} + \left(x - \frac{\pi}{2}\right) + y - \frac{\pi}{4}y^2$$

Mostly done well. Apart from arithmetic error, a few people didn't evaluate the derivatives at $(\pi/2,0)$ *and left terms of* $y\sin x$*, etc, in the answer. Others forgot to include* f_{xy}*.*

(ii) Setting $z_0 = f(\pi/2,0) = \dfrac{\pi}{2}$, the equation of the tangent plane at $(\pi/2,0)$ is given by

$$z - z_0 = f_x(\pi/2,0)\left(x - \frac{\pi}{2}\right) + f_y(\pi/2,0)\,(y-0) \Rightarrow z - \frac{\pi}{2} = \left(x - \frac{\pi}{2}\right) + y \Rightarrow x + y - z = 0\,.$$

Few people attempted this. Those that did got it mostly right, though quite a few people included the quadratic term in y.

(c) (i) Use the method of Lagrange multipliers to find the point on the plane with cartesian equation

$$5x+y-4z=11$$

which is nearest to the point $(3,1,-4)$. [5]

[*Hint: a distance function and its square have the same minimum.*]

Solution: We need a point on the plane (x,y,z) which is nearest to $(3,1,-4))$, so the distance between these is the function to be minimized. Using the hint, we see that when the distance D is minimal, so is its square $D^2=f$

$$D(x,y,z)=\sqrt{(x-3)^2+(y-1)^2+(z+4)^2} \Rightarrow f(x,y,z)=(x-3)^2+(y-1)^2+(z+4)^2$$

and getting rid of the root makes it easier to minimize $f(x,y,z)$. This is subject to the constraint that the point is on the plane, giving the constraint function

$$g(x,y,z)=5x+y-4z-11=0\ .$$

We set
$\nabla f=\lambda\nabla g \Rightarrow$
$f_x=2(x-3)=\lambda g_x=5\lambda \ \Rightarrow x=3+\frac{5}{2}\lambda$,
$f_y=2(y-1)=\lambda g_y=\lambda \ \Rightarrow y=1+\frac{1}{2}\lambda$,
$f_z=2(z+4)=\lambda g_z=-4\lambda \ \ z=-4-2\lambda$,
Substitute these into the constraint:

$$5x+y-4z \ = \ 5\left(3+\frac{5}{2}\lambda\right)+1+\frac{1}{2}\lambda-4(-4-2\lambda)=11 \ \Rightarrow \ \lambda=-1\ .$$

Finally, we have the point on the plane, given by $x=3+\frac{5}{2}\lambda=\frac{1}{2}$ $y=1+\frac{1}{2}\lambda=\frac{1}{2}$ and $z=-4-2\lambda=-2$. The point on the plane nearest to $(3,1,-4)$ is $(1/2,1/2,-2)$.

A lot of people didn't even attempt this. Of those that did, apart from minor arithmetic messups, the vast majority got it right.

(ii) Confirm the result from (i) by finding the intersection of the plane and a line with vector equation $\underline{\mathbf{r}}=\underline{\mathbf{a}}+\mu\underline{\mathbf{b}}$, where μ is any real number, and $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ need to be found. [4]

Solution: To confirm this using the intersection of a line with the plane, we require the line through $(3,1,-4)$, orthogonal to the plane. The line has equation

$$\underline{\mathbf{r}}=(3,1,-4)^T+\mu(5,1,-4)^T$$

Hence we need $x=3+5\mu,\ y=1+\mu,\ z=-4-4\mu$. To find the intersection, we substitute these into the plane equation and find the required value of μ:

$$5(3+5\mu)+1+\mu-4(-4-4\mu)=11 \ \Rightarrow \mu=-\frac{1}{2}$$

. Substituting this value into the equation of the line we get the same point.

Most people got this, though some used projection. Quite a few people calculated the distance between point and plane, correctly, though this was not asked for.

3. (a) Three vectors in $\mathbb{R}^3$ are given as $\vec{x} = (1,0,3)^T$,$\vec{y} = (2,1,4)^T$, and $\vec{z} = (-1,1,0)^T$, the position vectors of points X, Y and Z. A line L goes through X and Y. Let the line have direction $\underline{\mathbf{a}}$. Find the projection of the vector $\overrightarrow{XZ}$ onto $\underline{\mathbf{a}}$. Hence, or otherwise, find the point Q on line L which is nearest to Z, and find the shortest distance from Z to L. Explain your reasoning and provide a sketch! [7]

Solution: (i) The direction vector for L can be taken as $\overrightarrow{XY} = \underline{\mathbf{a}} = (1,1,1)^T$. We need $\overrightarrow{XZ} = (-2,1,-3)^T$, so the projection of $\overrightarrow{XZ}$ onto $\underline{\mathbf{a}}$ is given by $\lambda\underline{\mathbf{a}}$, where

$$\lambda = \frac{\underline{\mathbf{a}} \cdot \overrightarrow{XZ}}{\underline{\mathbf{a}} \cdot \underline{\mathbf{a}}} = \frac{-2+1-3}{1+1+1} = -\frac{4}{3} .$$

So the projection is $-\frac{4}{3}(1,1,1)^T$, see sketch, at end.

Taking this as a vector along L, beginning at X, we have a right-angled triangle XQZ, where $\overrightarrow{XQ}$ is the projection of $\overrightarrow{XZ}$ onto L, and the right-angle on L is at the point Q, nearest to Z, given that $\overrightarrow{QZ}$ is perpendicular to the line. So if $\overrightarrow{XQ} = -\frac{4}{3}(1,1,1)^T$, we can deduce that Q has coordinates $\frac{1}{3}(-1,-4,5)^T$ and the shortest distance from Z to the line is given by

$$|\overrightarrow{ZQ}| = |\frac{1}{3}(2,-7,5)^T| = \frac{1}{3}\sqrt{78} .$$

A lot of good answers. Many people didn't read/follow the instructions and just implemented a distance formula from high school, not always correctly. Many that found the correct distance this way, then forgot to find Q. Many forgot the sketch.

(b) A double integral is given by

$$I \;=\; \int\int_R (x+y)^n \, f(x^2-y^2)\, dx\, dy \;,$$

where R is the region of the $xy-$plane enclosed by the coordinate axes and the line $y = 1-x$.

(i) Use the change of variables $u = x^2-y^2$, $v = x+y$ to rewrite the integral as

$$I \;=\; \frac{1}{2}\int\int_{R*} v^{n-1} f(u)\, du\, dv \;,$$

and sketch the region R^* in the $uv-$plane.

(ii) Rewrite the double integral in two ways: once integrating first with respect to u, the other integrating first with respect to v.

(iii) Finally, evaluate the integral for the case $n = 2$ and $f(u) = e^u$. You can choose to integrate first with respect to u or v, but one of these will be easier! [10]

Solution: First, find $R*$. Clearly $x+y = 1 \Rightarrow v = 1$. Letting $x = 0$ we get $u = -y^2, v = y \Rightarrow u = -v^2$, and similarly $y = 0 \Rightarrow u = v^2$. Sketch of R^* at end.

(i) The Jacobian can be obtained easily as

$$J_{x,y}(u,v) \;=\; \frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x} \;=\; 2x(1) - (-2y)(1) = 2(x+y) = 2v \Rightarrow J_{u,v}(x,y) = \frac{1}{2v}$$

so $dxdy = \dfrac{1}{2v} dudv$ and

$$I \;=\; \int\int_{R^*} v^n f(u) \frac{1}{|2v|} du\, dv \;.$$

(ii) Given R^*, it's clearly easier to integrate first w.r.t u. Let $u = -v^2 \ldots v^2$ and then $v = 0 \ldots 1$. So $|v| = v$, and we can write

$$I \;=\; \frac{1}{2}\int_{v=0}^{1} \left(\int_{u=-v^2}^{v^2} v^{n-1} f(u)\, du \right) dv \;.$$

Integrating first w.r.t v, we have to split the region in two parts and obtain

$$I \;=\; \frac{1}{2}\int_{u=-1}^{0} \left(\int_{v=\sqrt{-u}}^{1} v^{n-1} f(u)\, dv \right) du \;+\; \frac{1}{2}\int_{u=0}^{1} \left(\int_{v=\sqrt{u}}^{1} v^{n-1} f(u)\, dv \right) du$$

(iii) We integrate first w.r.t u. so letting $f(u) = e^u$ and $n = 2$, we have the inner integral

$$\int_{u=-v^2}^{v^2} ve^u \, du \;=\; ve^u]_{u=-v^2}^{v^2} \;=\; ve^{v^2} - ve^{-v^2} \;.$$

and integrating w.r.t v we get

$$I \;=\; \frac{1}{4}\int_0^1 2ve^{v^2} - 2ve^{-v^2}\, dv \;=\; \frac{1}{4}\left[e^{v^2} + e^{-v^2}\right]_0^1 \;=\; \frac{1}{4}\left(e + \frac{1}{e} - 2\right) \;.$$

The most difficult question on the exam, with the fewest correct answers. Almost everyone got the Jacobian in (i) correctly, but not the sketch of R^*. For (ii,iii), most people who tried this, took R^* as a rectangle or a triangle, where the latter required more work.

(c) If A is a 4×3 matrix, what can you say about rank, pivot and free columns, nullspace and column space of A? In each case, what does this imply for the solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$? Assume $\underline{\mathbf{b}}$ is not the zero vector. When solutions exist, give a geometric description. [8]

Solution: Regardless of the rank, solutions to $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ exist if $\underline{\mathbf{b}}$ is in the column space, otherwise there will be no solutions.

The matrix has three columns, so the rank will be $r \leq 3$, as each column can have at most one pivot. If $r = 3$, we have three independent pivot columns, and no free columns. In that case the nullspace $N(A)$ will have only one element, the zero vector. The column space $C(A)$, as the linear combinations of the pivot columns, will be a 3-dimensional subspace of $\mathbb{R}^4$. If a solution to $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ exists, it will be a unique point in $\mathbb{R}^3$.

If $r = 2(\text{or} 1)$, there will be two (or one) pivot columns, and one (or two) free columns, giving one (or two) special solutions to $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$, and the Nullspace, formed as the linear combinations of the special solutions, will be a line (or a plane), through the origin, in $\mathbb{R}^3$. The column space will be a 2-D, (or 1-D) subspace of $\mathbb{R}^4$. If solutions exist, they will be a line (or a plane) in $\mathbb{R}^3$, parallel to the Nullspace.

Very many good answers. A lot of people stayed too general, with statements such as "the nullspace depends on the number of free variables" and "the column space depends on the rank", or "solutions are parallel to the nullspace." For an 8-mark question, a lot more is expected. On the other hand, some even thought about the $r = 0$ *case, very impressive!*

4. (a) Obtain constants α and β so the first-order Ordinary Differential Equation

$$\alpha xt + \cos x \cos t \; + \frac{1}{t} \; + \left(t^2 - x^2 + \beta \sin x \sin t \; \right) \frac{dx}{dt} = 0$$

is exact. Hence solve the equation subject to the initial condition $x = 0$ when $t = \pi$. [7]

Solution: Identify $P(t,x) = \alpha xt + \cos x \cos t \; + \frac{1}{t}$, and $Q(t,x) = t^2 - x^2 + \beta \sin x \sin t$, as in the formula sheet. For this to be an exact ODE, we require

$$\frac{\partial P}{\partial x} \; = \; \alpha t - \sin x \cos t \; = \; \frac{\partial Q}{\partial t} \; = \; 2t + \beta \sin x \cos t \; ,$$

The two are equal if $\alpha = 2$ and $\beta = -1$. With these values, the ODE is exact, and a function $f(t,x)$ exists with

$$\frac{\partial f}{\partial t} \; = \; P(t,x) \; = \; 2xt + \cos x \cos t + \frac{1}{t} \; , \quad \text{and} \quad \frac{\partial f}{\partial x} \; = \; Q(t,x) \; = \; t^2 - x^2 - \sin x \sin t \; ,$$

and we can integrate to find f:

$$f(t,x) \; = \; \int P \, dt \; = \; \int 2xt + \cos x \cos t + \frac{1}{t} \, dt \; = \; xt^2 + \cos x \sin t + \ln t + g(x) \; ,$$

and

$$f(t,x) \; = \; \int Q \, dx \; = \; \int t^2 - x^2 - \sin x \sin t \, dx \; = \; xt^2 - \frac{1}{3}x^3 + \cos x \sin t + h(t) \; ,$$

where $g(x)$ and $h(t)$ are arbitrary functions. Comparing the two expressions, we see that $h(t) = \ln t$, and $g(x) = -\frac{1}{3}x^3$ and the solution is given by $f(t,x) = constant$:

$$xt^2 - \frac{1}{3}x^3 + \cos x \sin t + \ln t = \; C \; ,$$

and we apply the initial condition $x(\pi) = 0$ to find $C = \ln \pi$.

Almost everyone got this. A few people mixed up P and Q, getting odd results. Others left the answer in the incorrect form $f(x,t) = xt^2 - \frac{1}{3}x^3 + \cos x \sin t + \ln t + C$, *others forgot to find C.*

(b) Use a Fourier series to show that, for $0 < x < \pi$,

$$\sin x = \frac{2}{\pi} - \frac{4}{\pi}\left(\frac{1}{1\cdot 3}\cos(2x) + \frac{1}{3\cdot 5}\cos(4x) + \frac{1}{5\cdot 7}\cos(6x) + \frac{1}{7\cdot 9}\cos(8x) + \cdots\right) . \qquad [\ 10\ ]$$

You may assume the identity $\sin[(m-n)x] + \sin[(m+n)x] = 2\sin(mx)\cos(nx)$.

Solution: We are given cosine terms, so need an even extension, so $b_n = 0$ and we only find a_n. Let $f(x) = |\sin x|$. We can now choose the interval for the period $T = 2\pi$ or $T = \pi$, given the symmetry.

With $T = \pi$, using the formula sheet:

$$a_n \;=\; \frac{2}{\pi}\int_{-\pi/2}^{\pi/2} f(x)\cos(2nx)\,dx \;=\; \frac{4}{\pi}\int_0^{\pi/2} \sin x\cos(2nx)\,dx \;=\; \frac{2}{\pi}\int_0^{\pi/2} \sin(1-2n)x + \sin(1+2n)x\,dx$$

given that $f(x)$ is even for the first step, and for the second step we used the identity with $m = 1$. Integrate to obtain

$$a_n \;=\; -\frac{2}{\pi}\left[\frac{\cos(1-2n)x}{1-2n} + \frac{\cos(1+2n)x}{1+2n}\right]_0^{\pi/2}$$

which simplifies nicely as the upper limit gives $\cos(k\pi/2) = 0$ for odd k, leaving only

$$a_n = \frac{2}{\pi}\left(\frac{1}{1-2n} + \frac{1}{1+2n}\right) \;=\; -\frac{4}{\pi}\frac{1}{(2n-1)(2n+1)} .$$

For the constant term:

$$a_0 = \frac{4}{\pi}\int_0^{\pi/2} \sin x\,dx \;=\; \left[-\frac{4}{\pi}\cos x\right]_0^{\pi/2} \;=\; \frac{4}{\pi} .$$

Hence, for $0 \le x \le \pi$ we have

$$f(x) \;=\; \sin x \;= \frac{2}{\pi} - \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)}\cos(2nx) ,$$

and letting $n = 1, 2, 3\ldots$, we obtain the series given in the question.

The choice $T = 2\pi$ makes the process a bit longer. Using the formula sheet, we have

$$a_n \;=\; \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)\,dx \;=\; \frac{2}{\pi}\int_0^{\pi} \sin x\cos(nx)\,dx \;=\; \frac{1}{\pi}\int_0^{\pi} \sin[(1+n)x] + \sin[(1-n)x]\,dx$$

given that $f(x)$ is even for the first step, and for the second step we used the identity with $m = 1$. Hence, for $n \neq 1$, we obtain

$$a_n \;=\; -\frac{1}{\pi}\left[\frac{\cos(1+n)x}{1+n} + \frac{\cos(1-n)x}{1-n}\right]_0^{\pi} \;=\; -\frac{1}{\pi}\left[\frac{\cos[(1+n)\pi]-1}{1+n} + \frac{\cos[(1-n)\pi]-1}{1-n}\right]$$

For odd n, $1 \pm n$ is even, so $\cos[(1\pm n)\pi] - 1 = 0$, and for even n, $1 \pm n$ is odd, so $\cos[(1\pm n)\pi] - 1 = -2$ and we get

$$a_n \;=\; \frac{2}{\pi}\left(\frac{1}{1+n} + \frac{1}{1-n}\right) = -\frac{4}{\pi(n^2-1)} \;=\; -\frac{4}{\pi(n-1)(n+1)}$$

for even n and zero for odd n. The above required $n \neq 1$, which we do separately:

$$a_1 = \frac{1}{\pi}\int_0^{\pi} \sin 2x + 0 \, dx \;=\; -\frac{1}{\pi}\left[\frac{1}{2}\cos 2x\right]_0^{\pi} \;=\; -\frac{1}{2\pi}(\cos 2\pi - 1) \;=\; 0\,.$$

Likewise,

$$a_0 = \frac{2}{\pi}\int_0^{\pi} \sin x \, dx \;=\; \left[-\frac{2}{\pi}\cos x\right]_0^{\pi} \;=\; -\frac{2}{\pi}(\cos \pi - 1) \;=\; \frac{4}{\pi}\,.$$

Finally, we have that, when $0 < x < \pi$,

$$f(x) \;=\; \sin x \;= \frac{2}{\pi} - \frac{4}{\pi}\sum_{even\ n}^{\infty} \frac{1}{(n-1)(n+1)}\cos(nx)\,,$$

and letting $n = 2,4,6,8\ldots$, we obtain the series given in the question.

Another tough question. A lot of people tried finding b_n because $\sin x$ is odd. Many people tried integrating from $-\pi$ to π or from $-\pi/2$ to $\pi/2$. Either way, this usually led to arithmetic error. Some people using $T = \pi$ integrated from from 0 to π, leading to difficulty. Of those that used $T = 2\pi$, noone checked the case a_1 separately, to exclude division by zero. The most common errors were getting to some version of

$$\left[\frac{\cos(1+kn)x}{1+kn} + \frac{\cos(1-kn)x}{1-kn}\right]_0^{\pi/k}$$

with $k = 1$ or $k = 2$, and then running into various forms of trouble. Many tried to force this to match the expected outcome, with varyingly large leaps of deduction.

(c) Let a matrix be given as

$$A = \begin{pmatrix} -1 & 2 & 1 & 0 \\ 0 & 1 & -2 & 3 \\ 2 & -3 & -4 & 3 \\ 0 & -2 & 4 & -6 \end{pmatrix}$$

Use Gaussian elimination to obtain the echelon form of A, identify rank, pivot and free columns. Hence find the nullspace $N(A)$ and column space $C(A)$ as the linear combinations of independent vectors. Finally, obtain a particular solution of $A\underline{\mathbf{x}} = \underline{\mathbf{b}} = (3,-4,-10,8)^T$, and with no further calculation, write down the general solution. [8]

Solution: Begin with Gaussian elimination, identifying the first pivot as $a_{11} = -1$, and including $\underline{\mathbf{b}}$ from the beginning: $(A : \underline{\mathbf{b}}) =$

$$\begin{pmatrix} \boxed{-1} & 2 & 1 & 0 & : & 3 \\ 0 & 1 & -2 & 3 & : & -4 \\ 2 & -3 & -4 & 3 & : & -10 \\ 0 & -2 & 4 & -6 & : & 8 \end{pmatrix} \begin{matrix} \\ \\ \\ R_3+2R_1 \end{matrix} \sim \begin{pmatrix} \boxed{-1} & 2 & 1 & 0 & : & 3 \\ 0 & 1 & -2 & 3 & : & -4 \\ 0 & -1 & -2 & 3 & : & -4 \\ 0 & -2 & 4 & -6 & : & 8 \end{pmatrix} \begin{matrix} \\ \\ R_3+R_2 \\ R_4+2R_2 \end{matrix} \sim \begin{pmatrix} \boxed{-1} & 2 & 1 & 0 & : & 3 \\ 0 & \boxed{1} & -2 & 3 & : & -4 \\ 0 & 0 & 0 & 0 & : & 0 \\ 0 & 0 & 0 & 0 & : & 0 \end{pmatrix}.$$

where the second pivot at $a_{22} = 1$ gives us the echelon form, after only three row operations. Pivot columns are therefore columns 1 and 2, columns 3 and 4 are free. The rank is the number of pivot columns, so $\text{rank}(A) = 2$.

In echelon form, pivot columns are independent, so the column space is formed by all linear combinations of the pivot columns:

$$C(A) = p(-1,0,2,0)^T + q(2,1,-3,-2)^T ,$$

for $p,q \in \mathbb{R}$.

To find the nullspace we solve $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$. Free variables are x_3 and x_4, and we begin by letting $x_3 = 1$ and $x_4 = 0$. In echelon form, we are left with two equations

$$-x_1 + 2x_2 + x_3 = 0 \quad \text{and} \quad x_2 - 2x_3 + 3x_4 = 0$$

From the second equation, $x_3 = 1, x_4 = 0 \Rightarrow x_2 = 2$. Substitute these into the first equation and get $x_1 = 5$, giving the first special solution $\underline{\mathbf{x}}_1$. Letting $x_3 = 0, x_4 = 1$, the second equation gives $x_2 = -3$ and substituting into the first equation, we get $x_1 = -6$, giving the second special solution $\underline{\mathbf{x}}_2$. The choice of x_3, x_4 makes $\underline{\mathbf{x}}_{1,2}$ independent, so the nullspace is all linear combinations of these:

$$N(A) = a\underline{\mathbf{x}}_1 + b\underline{\mathbf{x}}_2 = a\begin{pmatrix} 5 \\ 2 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} -6 \\ -3 \\ 0 \\ 1 \end{pmatrix},$$

for $a,b \in \mathbb{R}$.

Finally, considering the last column of the echelon form, we get two equations:

$$-x_1 + 2x_2 + x_3 = 3 \quad \text{and} \quad x_2 - 2x_3 + 3x_4 = -4$$

and setting the free variables $x_3 = x_4 = 0$, the second equation gives $x_2 = -4$, and hence, the first equation has $x_1 = -11$, giving a particular solution $\underline{\mathbf{x}}_p = (-11,-4,0,0)^T$ and the general solution is obtained by taking the particular solution and the nullspace:

$$\underline{\mathbf{x}} = \begin{pmatrix} -11 \\ -4 \\ 0 \\ 0 \end{pmatrix} + a\begin{pmatrix} 5 \\ 2 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} -6 \\ -3 \\ 0 \\ 1 \end{pmatrix}.$$

Mostly well done. A few people did Gaussian incorrectly getting rank 3, but then usually continuing correctly from there. A few people wrote the column space as the combinations of all four columns. Some wrote that

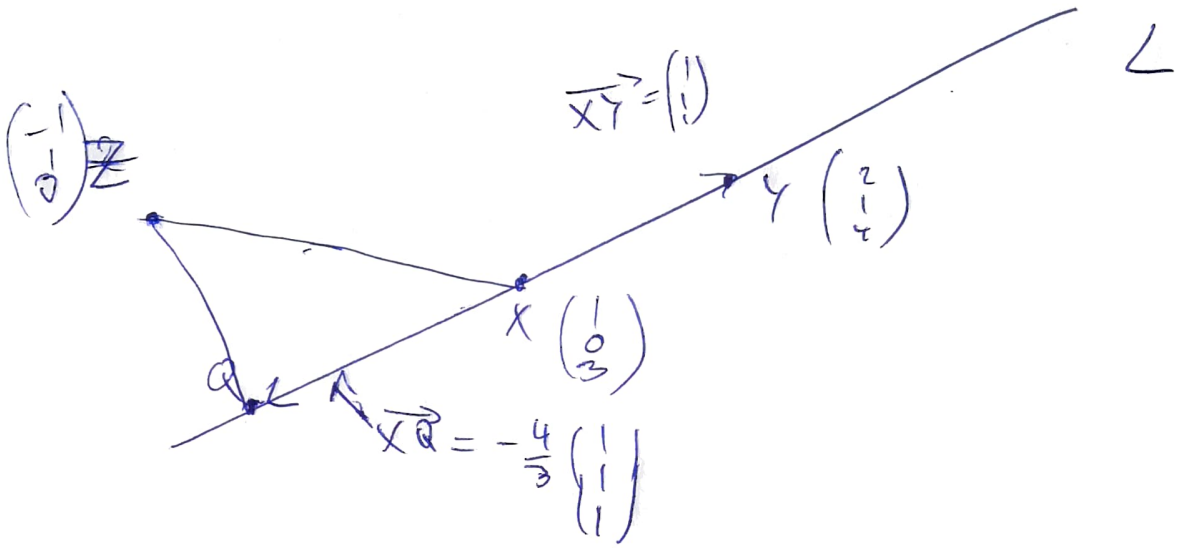
$$N(A) = \left\{ \begin{pmatrix} 5 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -6 \\ -3 \\ 0 \\ 1 \end{pmatrix} \right\},$$

which is a set containing only two vectors, and/or similarly for $C(A)$. Quite a few solved

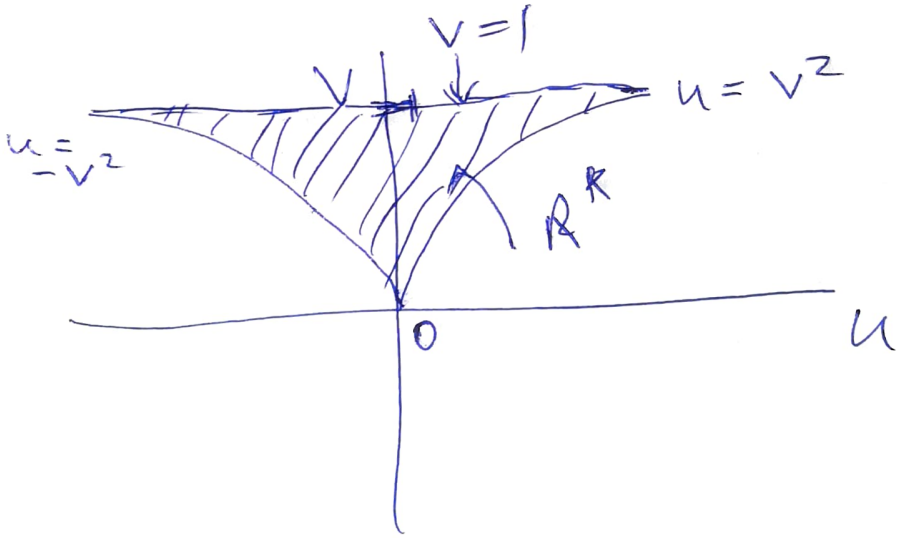
$$-x_1 + 2x_2 + x_3 = 3 \quad \text{and} \quad x_2 - 2x_3 + 3x_4 = -4$$

to find the nullspace.

Q 3a


L
$\overrightarrow{XY} = \begin{pmatrix}1\\1\\1\end{pmatrix}$
$\begin{pmatrix}-1\\-1\\0\end{pmatrix}$ Z
Y $\begin{pmatrix}2\\1\\4\end{pmatrix}$
X $\begin{pmatrix}1\\0\\3\end{pmatrix}$
Q
$\overrightarrow{XQ} = -\frac{4}{3}\begin{pmatrix}1\\1\\1\end{pmatrix}$

Q 3b


v = 1
V
$u = v^2$
$u = -v^2$
R
0
u